



A1	BONNYRIGG 2020	☒	☐
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Two forces, F_1 and F_2 , both measured in newtons, act on a mass of 3 kg, producing an acceleration of $\sqrt{3}\mathbf{i} + \mathbf{j} \text{ ms}^{-2}$. 1

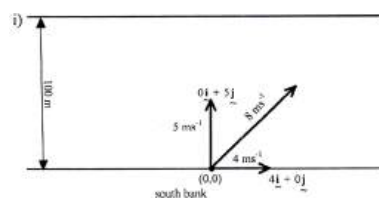
Given that $F_1 = 2\mathbf{j}$, the acute angle between F_1 and F_2 is

- A. $\arccos\left(\frac{1}{\sqrt{3}}\right)$ D. $\arccos\left(\frac{1}{2\sqrt{7}}\right)$
 B. $\pi - \arccos\left(\frac{1}{\sqrt{3}}\right)$ E. $\pi - \arccos\left(\frac{1}{2\sqrt{7}}\right)$
 C. $\frac{\pi}{6}$

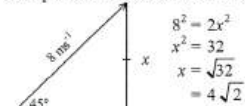
A2	CHELtenham GIRLS 2020	☒	☐
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Sienna intends to row her boat from the south bank of a river to meet with her friends on the north bank. The river is 100 metres wide. Sienna's rowing speed is 5 metres per second when the water is still. The river is flowing east at a rate of 4 metres per second. Sienna's boat is also being impacted by a wind blowing from the south-west, which pushed the boat at 8 metres per second. She starts rowing across the river by steering the boat such that it is perpendicular to the south bank.

- i) Show that the velocity of Sienna's boat can be expressed as the component vector: 2
 $(4 + 4\sqrt{2})\mathbf{i} + (5 + 4\sqrt{2})\mathbf{j}$
 ii) Calculate the speed of the boat, correct to 2 decimal places. 1
 iii) Determine the distance rowed from Sienna's starting point to her landing point and how long it will take her to reach the north bank. 2



Component vectors of the wind



Wind vector is $4\sqrt{2}\mathbf{i} + 4\sqrt{2}\mathbf{j}$

Rowing vector is $0\mathbf{i} + 5\mathbf{j}$

Water flow vector is $4\mathbf{i} + 0\mathbf{j}$

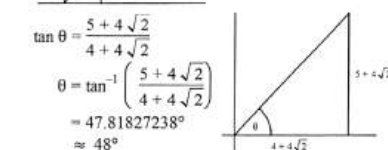
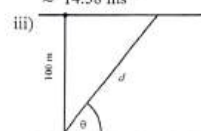
Resultant velocity = $4\mathbf{i} + 0\mathbf{j} + 0\mathbf{i} + 5\mathbf{j} + 4\sqrt{2}\mathbf{i} + 4\sqrt{2}\mathbf{j}$
 $= (4 + 4\sqrt{2})\mathbf{i} + (5 + 4\sqrt{2})\mathbf{j}$

ii) The velocity is equal to the magnitude of the resultant velocity vector from (a).

$$v = \sqrt{(4 + 4\sqrt{2})^2 + (5 + 4\sqrt{2})^2}$$

$$= 14.38135517$$

$$\approx 14.38 \text{ ms}^{-1}$$



The distance from start to finish is given by d .

$$\sin \theta = \frac{100}{d}$$

$$d = \frac{100}{\sin 48^\circ} = 134.9493465 \text{ m} \approx 135 \text{ m}$$

$$\text{Time taken} = \frac{134.9493465 \text{ m}}{14.38 \text{ ms}^{-1}}$$

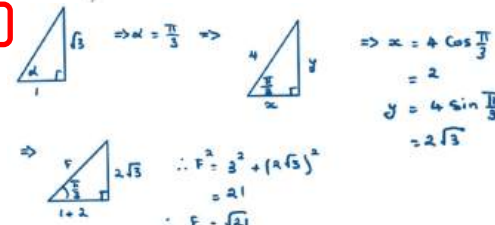
$$= 9.384516448 \approx 9.4 \text{ seconds}$$

A3	FORT STREET 2020	☒	☐
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Two forces P and Q act on a body. P acts in the direction $\mathbf{i}$ with magnitude one newton and Q acts in the direction of $\mathbf{i} + \sqrt{3}\mathbf{j}$ with magnitude four newtons. 1

The magnitude of the total force acting on the body, in newtons, is

- A. 3 B. $\sqrt{15}$ C. $\sqrt{17}$ D. $\sqrt{21}$



A4	HUNTERS HILL 2020	☒	☐
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The work done, W , by a constant force, F , in moving a particle through a displacement, s , is defined by the formula $W = F \cdot s$. A force described by the vector $F = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$ moves a particle along the line l from $P(-1, 2)$ to $Q(2, -2)$.

- (i) Find $s = \overrightarrow{PQ}$ and hence find the value of W . 1
 (ii) Hence, verify that W is also given by $W = (F \cdot \hat{s})|s|$. 1
 (iii) Find the component of F in the direction of l . 2

- (i) Substituting $F = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$ and $s = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$ into $W = F \cdot s$ gives:
 $W = F \cdot s = \begin{pmatrix} 4 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -4 \end{pmatrix} = 20$
 (ii) A unit vector in the direction of $\overrightarrow{PQ}$ is $\hat{s} = \frac{1}{5} \begin{pmatrix} 3 \\ -4 \end{pmatrix}$.
 Substituting $F = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$, $\hat{s} = \frac{1}{5} \begin{pmatrix} 3 \\ -4 \end{pmatrix}$ and $|s| = 5$ into $W = (F \cdot \hat{s})|s|$ gives:
 $W = \left(\begin{pmatrix} 4 \\ -2 \end{pmatrix} \cdot \frac{1}{5} \begin{pmatrix} 3 \\ -4 \end{pmatrix} \right) 5 = 20$
 (iii) The component of F in the direction of l is given by $(\frac{F \cdot \hat{s}}{\hat{s} \cdot \hat{s}})\hat{s}$.
 Substituting $F \cdot \hat{s} = 20$, $\hat{s} \cdot \hat{s} = 25$ and $\hat{s} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$ into $(\frac{F \cdot \hat{s}}{\hat{s} \cdot \hat{s}})\hat{s}$ gives:
 $(\frac{F \cdot \hat{s}}{\hat{s} \cdot \hat{s}})\hat{s} = \frac{20}{25} \begin{pmatrix} 3 \\ -4 \end{pmatrix} = \frac{4}{5} \begin{pmatrix} 3 \\ -4 \end{pmatrix} = \begin{pmatrix} 2.4 \\ -3.2 \end{pmatrix}$
 Alternatively, the component of F in the direction of l is $(F \cdot \hat{s})\hat{s}$.

A5	NORMANHURST BOYS 2020	☒	☐
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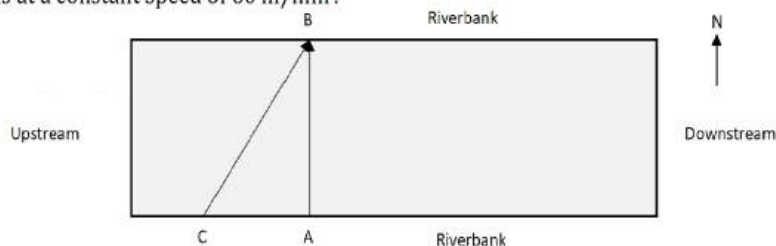
A force, described by the vector $F = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$, moves a particle along the line l from $A(-1, -1)$ to $B(2, -3)$.

- i. $\overrightarrow{AB} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ ii. Applying the projection of F on $\overrightarrow{AB}$, such that
 i. Find the unit vector that is in the direction of $\overrightarrow{AB}$. $|\overrightarrow{AB}| = \sqrt{9 + 4} = \sqrt{13}$ $\text{proj}_{\overrightarrow{AB}} F = \frac{F \cdot \overrightarrow{AB}}{|\overrightarrow{AB}|^2} \overrightarrow{AB}$
 ii. Find the component of F in the direction of line l . $\therefore$ unit vector is $\frac{1}{\sqrt{13}} \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ $\text{proj}_{\overrightarrow{AB}} F = \frac{F \cdot \overrightarrow{AB}}{|\overrightarrow{AB}|^2} \overrightarrow{AB}$
 $= \frac{\begin{pmatrix} 2 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -2 \end{pmatrix}}{13} \begin{pmatrix} 3 \\ -2 \end{pmatrix}$
 $= \frac{6 - 10}{13} \begin{pmatrix} 3 \\ -2 \end{pmatrix} = -\frac{4}{13} \begin{pmatrix} 3 \\ -2 \end{pmatrix}$



A6	JAMES RUSE 2020	☒	☐
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Danny wants to swim across the river from position A to position B shown on the diagram below. The riverbanks are parallel straight lines, and $\overline{AB}$ is perpendicular to the riverbanks. The speed of the river flow is 30 m/min from west to east and the river is 240 metres wide. Assume Danny swims at a constant speed of 60 m/min.



- In which direction should Danny aim to swim, in order to move from point A to point B in a straight line under the influence of the current? Draw a vector diagram to demonstrate the velocities and show all necessary working out.
- Gabriel also swims at 60 m/min. but he is to start at position C , which is 60 metres upstream from position A as shown in the diagram. In which direction should Gabriel aim to swim in order to swim directly to position B under the influence of the current? (Draw a vector diagram to demonstrate the velocities and show all necessary working out. Round your answer to one decimal place.)

- (i) Current is moving West to East, hence there'll be a component of velocity in the direction parallel to the bank.
If Danny is to go from A to B , he must head **into** the current, so that velocities reconcile as:

If t minutes is the time taken to make the journey, then we should have for displacement:



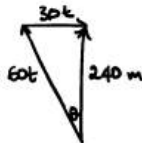
DB: $v_{\text{current}} \times t = 30t$ metres

AD: $v_{\text{Danny}} \times t = 60t$ metres

And we're given that AB is 240 metres. Hence

$$\sin \theta = \frac{30t}{60t} = \frac{1}{2} \rightarrow \theta = 30^\circ$$

So, Danny must swim N30°W.



- (ii) Gabriel swims at 60 m/min and starts at position C , to the West of A . By similar reasoning to (i), if the journey is completed in t minutes, we want:

where $tv_{\text{current}} = 30t$ and $tv_{\text{Gabriel}} = 60t$.

We need $\angle BCE + \angle BCA \equiv \theta + \psi$.

From the construction, $\tan \psi = \frac{240}{60} = 4$. So,
 $\psi = \tan^{-1} 4 \approx 76^\circ$

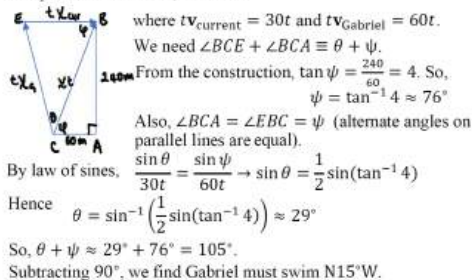
Also, $\angle BCA = \angle ECB = \psi$ (alternate angles on parallel lines are equal).

$\frac{\sin \theta}{30t} = \frac{\sin \psi}{60t} \rightarrow \sin \theta = \frac{1}{2} \sin(\tan^{-1} 4)$

Hence $\theta = \sin^{-1} \left(\frac{1}{2} \sin(\tan^{-1} 4) \right) \approx 29^\circ$

So, $\theta + \psi \approx 29^\circ + 76^\circ = 105^\circ$.

Subtracting 90° , we find Gabriel must swim N15°W.

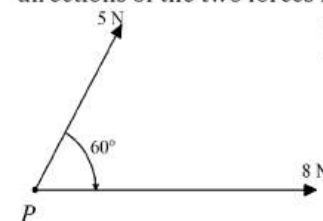


A7	ST GEORGE GIRLS 2020	☒	☐
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Forces of magnitude 8 N and 5 N act on a particle P . The angle between the directions of the two forces is 60° as shown in the diagram.

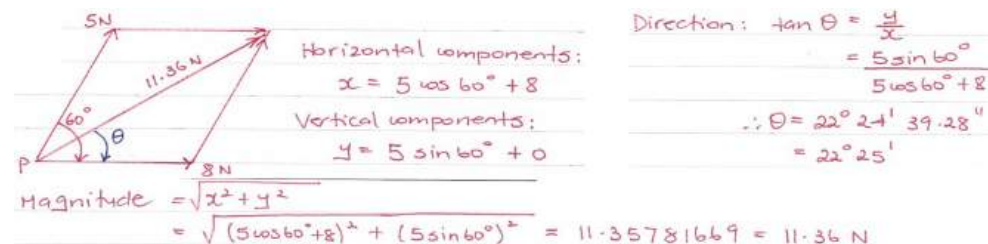
Which of the following is the correct magnitude and direction of the resultant force acting on P ?

- 11.36 N, $22^\circ 25'$ to the horizontal
- 11.36 N, $67^\circ 35'$ to the horizontal
- 12.58 N, $22^\circ 25'$ to the horizontal
- 12.58 N, $67^\circ 35'$ to the horizontal



1

3



A8	SYDNEY GIRLS 2020	☒	☐
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The movements of a drone flying straight and level at a constant speed are being monitored via a GPS screen.

One unit on the screen represents 100 m in the air.

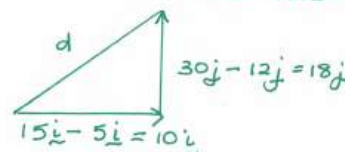
The initial position vector of the drone on the screen is

$$5\mathbf{i} + 12\mathbf{j}. \text{ An hour later its position vector on the screen is } 15\mathbf{i} + 30\mathbf{j}.$$

Find the distance travelled by the drone in km, correct to two decimal places.



$$\text{Speed} = \frac{\text{distance travelled}}{\text{time taken}}$$



$$\begin{aligned} 15\mathbf{i} - 5\mathbf{i} &= 10\mathbf{i} \\ 30\mathbf{j} - 12\mathbf{j} &= 18\mathbf{j} \\ \therefore d^2 &= 10^2 + 18^2 \\ d &= \sqrt{424} \text{ units} \times 100\text{m} = 2059\text{m} \\ \therefore \text{distance} &= 2.06 \text{ km (2 d.p.)} \end{aligned}$$



A9	PENRITH 2024	☒	☐
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Three forces $F_1 = 80 \text{ N}$ at 290° , $F_2 = 110 \text{ N}$ at 040° and $F_3 = 90 \text{ N}$ at 180° are acting on an object. Show that the resultant force $\vec{F}$ is $-4.5\hat{i} + 21.6\hat{j}$, corrected to one decimal place.

$$\vec{F}_1 = -80 \cos 20^\circ \hat{i} + 80 \sin 20^\circ \hat{j}$$

$$\vec{F}_2 = 110 \cos 50^\circ \hat{i} + 110 \sin 50^\circ \hat{j}$$

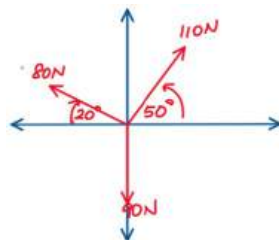
$$\vec{F}_3 = -90 \hat{j}$$

$$\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3$$

$$= (-80 \cos 20^\circ + 110 \cos 50^\circ) \hat{i} + (80 \sin 20^\circ + 110 \sin 50^\circ - 90) \hat{j}$$

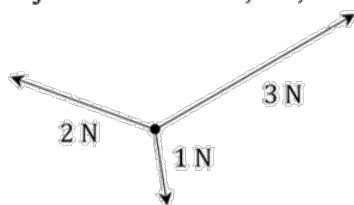
$$= -4.5 \hat{i} + 21.6 \hat{j}$$

Not well done.
Most students mixed sin and cos
and used 70° or 40° .



A10	CARINGBAH 2023	☒	☐
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A particle is subject to forces of 1 N, 2 N, and 3 N in the directions shown.



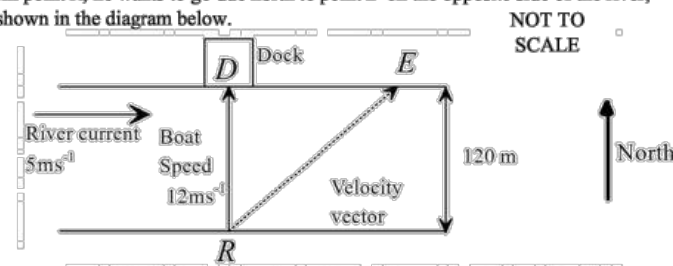
The sum of the forces acting on the particle is called the resultant force. Which diagram best shows the resultant force, R ?

- (A)
- (B)
- (C)
- (D)

A11	ABBOTSLEIGH 2023	☒	☐
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Rylie has a boat which moves at a top speed of 12 ms^{-1} in still water.

From point R , he wants to go due north to point D on the opposite side of the river, as shown in the diagram below.

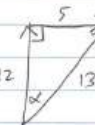


Today the current in the river is flowing at 5 ms^{-1} . From R , he steers the boat due north toward D at top speed. Due to the current he drifts down the river and arrives at point E .

- Taking R as the origin, write down Rylie's velocity vector in the form $x\hat{i} + y\hat{j}$ and find the magnitude of this vector.
- What is the bearing of Rylie's velocity vector and how far does he travel from R to E ?
- On what bearing should Rylie have pointed the boat, so that he arrived at D , with the boat travelling at top speed?

$$(i) \vec{v} = 5\hat{i} + 12\hat{j}$$

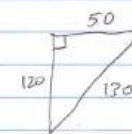
Using Pythagoras, $|\vec{v}| = \sqrt{5^2 + 12^2}$



$$(ii) \text{ From the diagram, } \tan \alpha = \frac{5}{12}$$

$$\therefore \alpha = 20.61^\circ$$

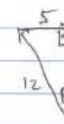
$\therefore$ Rylie's bearing is 022.6° and he travels 130 m .



(iii) using the triangle:

$$\sin \theta = \frac{5}{12}$$

$$\therefore \theta = 24.62^\circ$$



$\therefore$ Rylie should use a bearing of $360 - 24.62 = 335.38^\circ$



A12	HORNSBY GIRLS 2023	☒	☐
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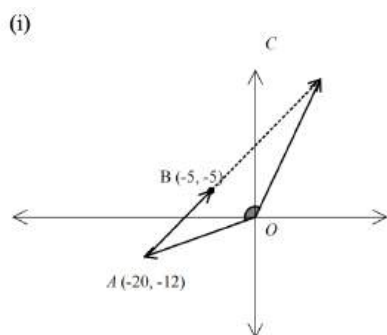
The position of a boat, relative to a stationary observer at the origin, point O , is being recorded.

The boat travels at a constant speed and in a constant direction and passes through three collinear points A , B and C . The boat is first observed at point A on a position vector of $-20\mathbf{i} - 12\mathbf{j}$.

One minute later, the boat is observed at point B on a position vector of $-5\mathbf{i} + 5\mathbf{j}$.

Using binoculars and without taking their eyes off the boat, the observer watches the motion of the boat for five minutes, starting when the boat is at point A , passing through the point B and continuing through to the point C .

- (i) Find the position vector of the boat after the five minutes when it is at point C . 2
- (ii) Hence, find $\angle AOC$, the angle through which the observer's binoculars move during this five minute interval, correct to the nearest degree. 2



$$\begin{aligned}\overrightarrow{AB} &= \begin{pmatrix} -5 \\ 5 \end{pmatrix} - \begin{pmatrix} -20 \\ -12 \end{pmatrix} = \begin{pmatrix} 15 \\ 17 \end{pmatrix} \\ \overrightarrow{OC} &= \overrightarrow{OA} + 5 \times \overrightarrow{AB} \\ &= \begin{pmatrix} -20 \\ -12 \end{pmatrix} + 5 \begin{pmatrix} 15 \\ 17 \end{pmatrix} \\ &= \begin{pmatrix} 55 \\ 73 \end{pmatrix}\end{aligned}$$

Hence, the position vector after 5 seconds is $55\mathbf{i} + 73\mathbf{j}$.

$$\begin{aligned}\text{(ii)} \quad \overrightarrow{OA} \cdot \overrightarrow{OC} &= |\overrightarrow{OA}| |\overrightarrow{OC}| \cos \theta \\ \begin{pmatrix} -20 \\ -12 \end{pmatrix} \cdot \begin{pmatrix} 55 \\ 73 \end{pmatrix} &= \sqrt{(-20)^2 + (-12)^2} \sqrt{(55)^2 + (73)^2} \times \cos \theta \\ \cos \theta &= \frac{-20 \times 55 - 12 \times 73}{\sqrt{(-20)^2 + (-12)^2} \sqrt{(55)^2 + (73)^2}} \\ &\approx -0.926915 \dots \\ \theta &\approx 157^\circ 57' 32.5'' \\ &\approx 158^\circ \text{ to the nearest degree}\end{aligned}$$

A13	SYDNEY GIRLS 2023	☒	☐
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Two forces act simultaneously on a particle situated at the origin, as shown.

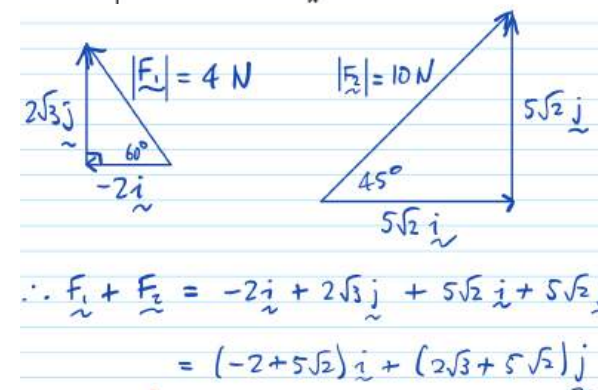
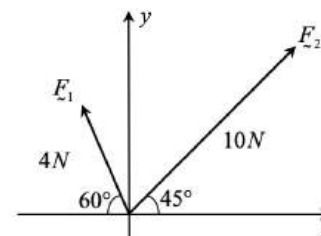
What is the resultant force, $\underline{F}_1 + \underline{F}_2$?

A. $(2 + 5\sqrt{2})\mathbf{i} + (-2\sqrt{3} + 5\sqrt{2})\mathbf{j}$

B. $(-2 + 5\sqrt{2})\mathbf{i} + (-2\sqrt{3} + 5\sqrt{2})\mathbf{j}$

C. $(-2 + 5\sqrt{2})\mathbf{i} + (2\sqrt{3} + 5\sqrt{2})\mathbf{j}$

D. $(2 + 5\sqrt{2})\mathbf{i} + (2\sqrt{3} + 5\sqrt{2})\mathbf{j}$



A14	JAMES RUSE 2023	☒	☐
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A plane experiences two forces: 10N of gravity straight down, and 60N of thrust at an angle of 15° above the horizontal. What is the magnitude of the net force to the nearest Newton?

(A) 50N

(B) 53N

(C) 58N

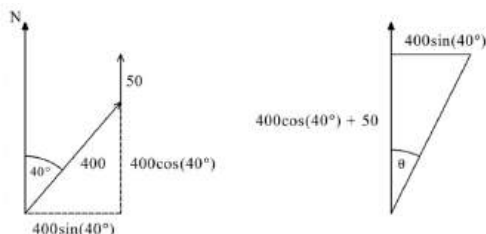
(D) 70N



A15	KILLARA 2023	☒	☐
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A plane is travelling at 400km/h with a bearing of 40° . The plane encounters a wind from the south with a speed of 50km/h. 2

What is the resultant bearing of the plane?



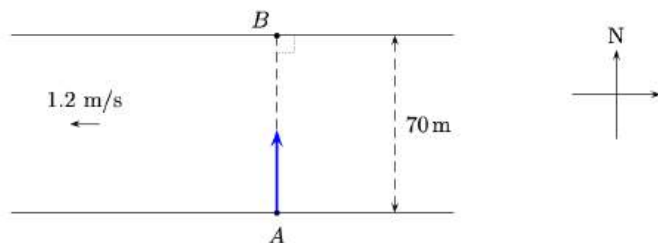
Using the diagram above, the resultant vector of the plane with wind is $\vec{v} = \begin{pmatrix} 400 \sin 40^\circ \\ 400 \cos 40^\circ + 50 \end{pmatrix}$

For the bearing:

$$\theta = \tan^{-1} \left(\frac{400 \sin 40^\circ}{400 \cos 40^\circ + 50} \right) \approx 036^\circ \text{T}$$

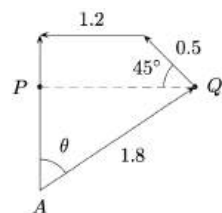
A16	NORMANHURST BOYS 2023	☒	☐
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A fish at point A wants to swim across the river that is 70 m wide. The banks of the river are parallel and the points A and B are on opposite sides of the river. 3



The fish swims at a speed of 1.8 m/s and the river is flowing downstream at 1.2 m/s. However, a 0.5 m/s wind blows constantly to the north-westerly direction.

What is the bearing that the fish should head to in order to land directly at point B from point A?



$$PQ = 1.2 + \frac{0.5}{\sqrt{2}} = \frac{6}{5} + \frac{1}{2\sqrt{2}}$$

In $\triangle APQ$,

$$\sin \theta = \frac{\frac{6}{5} + \frac{1}{2\sqrt{2}}}{1.8}$$

$$\theta = 59.6647\dots^\circ$$

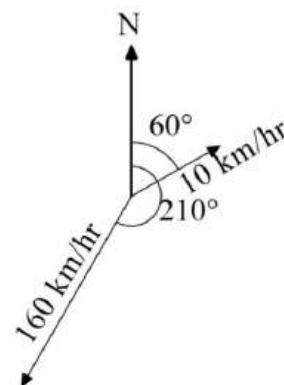
$\therefore$ The bearing that the fish should head to is 060° T (to the nearest degree) Page 5

A17	BOOKMINDERS 2022	☒	☐
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A light aircraft is travelling on a bearing of 210° at a speed of 160 km/hr. The aircraft is experiencing a wind with a speed of 10 km/hr on a bearing of 060° . 4

The aircraft needs to go to an island which is on a bearing of 210° and is 240 km from the origin.

Determine if the plane is within a 20 km radius of the island after 1.5 hours.



Horizontal:

$$v_x = 160 \cos(-120^\circ) + 10 \cos(30^\circ) = -71.3 \text{ km/hr}$$

$$v_y = 160 \sin(-120^\circ) + 10 \sin(30^\circ) = -133.6 \text{ km/hr}$$

Position:

$$x = -71.3 \times 1.5 = -107.0 \text{ km}$$

$$y = -133.6 \times 1.5 = -200.4 \text{ km}$$

Bearing:

$$90 - \tan^{-1} \frac{-200.4}{-107.0} + 180 = 208.1^\circ$$

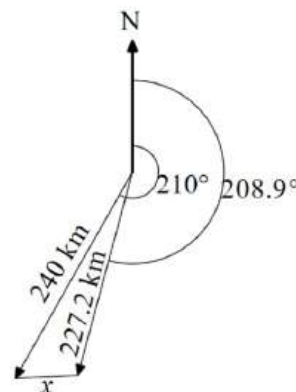
Distance:

$$\sqrt{107.0^2 + 200.4^2} = 227.2 \text{ km}$$

$$x^2 = 240^2 + 227.2^2 - 2(240)(227.2) \cos 1.1^\circ$$

$$x \approx 13.56 \text{ km}$$

The plane is within 20 km of the island.

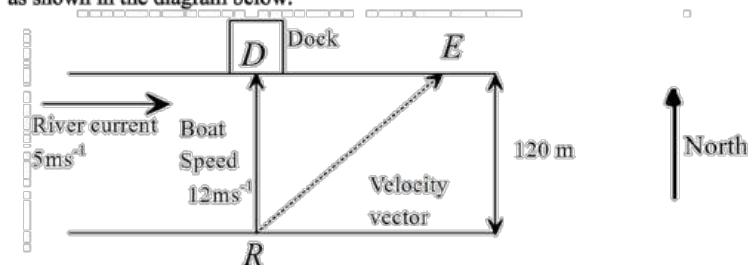




A18	ABBOTSLEIGH 2022	☒	☐
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Rylie has a boat which moves at a top speed of 12 ms^{-1} in still water.

From point R , he wants to go due north to point D on the opposite side of the river, as shown in the diagram below.



Today the current in the river is flowing at 5 ms^{-1} . From R , he steers the boat due north toward D at top speed. Due to the current he drifts down the river and arrives at point E .

- Taking R as the origin, write down Rylie's velocity vector in the form $x\mathbf{i} + y\mathbf{j}$ and find the magnitude of this vector.
- What is the bearing of Rylie's velocity vector and how far does he travel from R to E ?
- On what bearing should Rylie have pointed the boat, so that he arrived at D , with the boat travelling at top speed?

(i) $\vec{v} = 5\mathbf{i} + 12\mathbf{j}$

Using Pythagoras, $|\vec{v}| = \sqrt{5^2 + 12^2} = 13$

(ii) From the diagram, $\tan \alpha = \frac{5}{12}$

$\therefore \alpha = 22.61^\circ$

$\therefore$ Rylie's bearing is 022.6° and he travels 130m

(iii) using the triangle:

$\sin \theta = \frac{5}{12}$

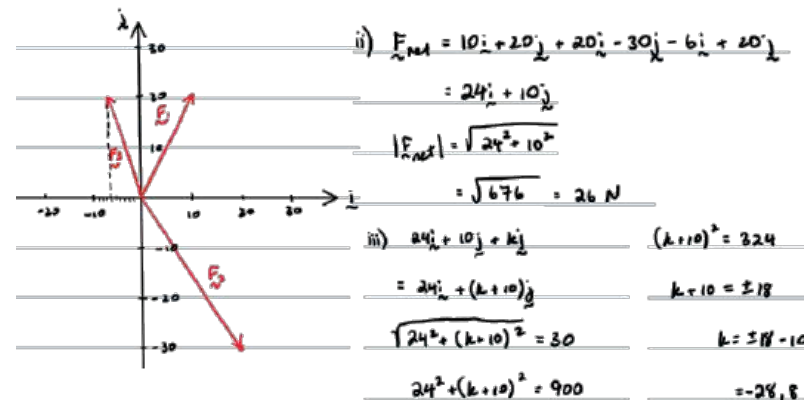
$\therefore \theta = 24.62^\circ$

ie Rylie should use a bearing of $360 - 24.62 = 335^\circ$

A19	JAMES RUSE 2022	☒	☐
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Three forces $\vec{F}_1$, $\vec{F}_2$, and $\vec{F}_3$ are represented by vectors $10\mathbf{i} + 20\mathbf{j}$, $20\mathbf{i} - 30\mathbf{j}$ and $-6\mathbf{i} + 20\mathbf{j}$ Newtons respectively.

- Draw a sketch showing the relative magnitude and direction of the three forces.
- Determine the resultant force, expressed as a vector, and find its magnitude.
- An additional force $\vec{F}_4 = k\mathbf{j}$ is now applied so that the resultant of $\vec{F}_1$, $\vec{F}_2$, and $\vec{F}_3$ and $\vec{F}_4$ has magnitude 30N . What is the value of k ?



A20	BONNYRIGG 2020	☒	☐
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A force described by the vector $\vec{F} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ newtons is applied to an object lying on a line ℓ which is parallel to the vector $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$.

- Find the component of the force $\vec{F}$ in the direction of the line ℓ .
- What is the component of the force $\vec{F}$ in the direction perpendicular to the line?

i. A unit vector in the direction of ℓ is $\hat{u} = \frac{1}{5} \begin{pmatrix} 3 \\ 4 \end{pmatrix}$

so the component of $\vec{F}$ in the direction of ℓ is

$(\vec{F} \cdot \hat{u}) \hat{u} = 2\hat{u} = \begin{pmatrix} 1.2 \\ 1.6 \end{pmatrix}$

ii. The component of $\vec{F}$ perpendicular to ℓ is

$\vec{F} - \begin{pmatrix} 1.2 \\ 1.6 \end{pmatrix} = \begin{pmatrix} 0.8 \\ -0.6 \end{pmatrix}$



A21	TOTAL EDUCATION 2020	☒	☐
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The position of a jet flying straight and level at a constant speed is plotted on its GPS screen. One unit on the screen represents 1 km in the air.

At 9:00am the position vector of the plane is $50\mathbf{i} + 12\mathbf{j}$. One minute later it is $35\mathbf{i} + 52\mathbf{j}$.

Find:

- The position vector at 9:20 am.
- The velocity (as a vector) in km/h.

(i) Jet's flight vector in 1st min

$$= (35\mathbf{i} + 52\mathbf{j}) - (50\mathbf{i} + 12\mathbf{j})$$

$$= -15\mathbf{i} + 40\mathbf{j}$$

in 20 min $\Rightarrow 20(-15\mathbf{i} + 40\mathbf{j})$

$$= -300\mathbf{i} + 800\mathbf{j}$$

(ii) Velocity vector $\frac{(-300\mathbf{i} + 800\mathbf{j})}{20}$

$$= -15\mathbf{i} + 40\mathbf{j} \text{ (20 min)}$$

$$= -750\mathbf{i} + 2400\mathbf{j} \text{ (1 hr)}$$

2
1

A22	HUNTERS HILL 2020	☒	☐
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A force described by the vector $\mathbf{F} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ newtons is applied to an object lying on the line l which is parallel to the vector $\begin{pmatrix} 2 \\ 5 \end{pmatrix}$

- Find the component of the force $\mathbf{F}$ in the direction of the line l .
- What is the component of the force perpendicular to the line l ?
- What force parallel to the line l must be applied to the object in order for it to remain stationary?

i) $\mathbf{F} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$

and vector in direction of l

$$\hat{\mathbf{s}} = \frac{1}{\sqrt{29}} \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

$\text{proj}_{\hat{\mathbf{s}}} \mathbf{F} = (\mathbf{F} \cdot \hat{\mathbf{s}}) \hat{\mathbf{s}}$

$$= \left(3 \cdot \frac{2}{\sqrt{29}} + (-2) \cdot \frac{5}{\sqrt{29}} \right) \cdot \frac{1}{\sqrt{29}} \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

$$= \frac{-4}{29} \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

ii) perpendicular component

$$= \mathbf{F} - \text{proj}_{\hat{\mathbf{s}}} \mathbf{F}$$

$$= \begin{bmatrix} 3 \\ -2 \end{bmatrix} - \frac{-4}{29} \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \frac{1}{29} \begin{bmatrix} 5 \\ 22 \end{bmatrix}$$

iii) opposite of parallel component force applied to object is

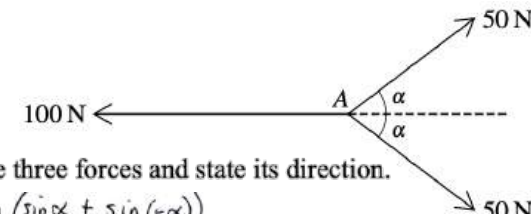
$$\frac{4}{29} \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

3
1
1

A23	SYDNEY BOYS 2022	☒	☐
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Three coplanar forces of magnitude 100 N, 50 N and 50 N act at a point A , as shown in the diagram below.

The angle α is such that $\cos \alpha = \frac{4}{5}$.



Find the magnitude of the resultant of the three forces and state its direction.

Vertical force components = $50\text{N}(\sin \alpha + \sin(-\alpha))$

$$= 0\text{N}$$

Horizontal force components = $50\text{N}(\cos \alpha + \cos(-\alpha)) - 100\text{N}$

$$= \frac{8}{5} \times 50\text{N} - 100\text{N}$$

$$= -20\text{N}$$

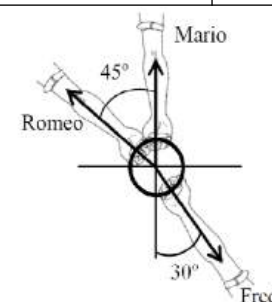
$\therefore$ The resultant is 20 N to the left.

A24	FORT STREET 2020	☒	☐
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Fred, Mario and Romeo are fighting over a crown.

The three of them are holding on to the crown in the formation as shown in the diagram.

If no one manages to pull the crown in their direction (ie, the crown does not move) and Fred is applying a force of 40 N, exactly how much force are Mario and Romeo applying to the crown?



Let $\mathbf{F}$, $\mathbf{M}$ and $\mathbf{R}$ be the force applied to the crown by Fred, Mario and Romeo respectively

$$\mathbf{F}_{\mathbf{i}} = 40 \sin 30^\circ, \quad \mathbf{F}_{\mathbf{j}} = -40 \cos 30^\circ$$

$$= 40 \times \frac{1}{2} = 20\text{ N}$$

$$= -40 \times \frac{\sqrt{3}}{2} = -20\sqrt{3}\text{ N}$$

Now $\mathbf{R}_{\mathbf{i}} = -\mathbf{F}_{\mathbf{i}}$ and $\mathbf{R}_{\mathbf{j}} = \mathbf{F}_{\mathbf{j}}$

$$= -20\text{ N}$$

$$= 20\text{ N}$$

$$\therefore \mathbf{R} = \sqrt{(-20)^2 + (20)^2}$$

$$= 20\sqrt{2}\text{ N}$$

Now $\mathbf{M} + \mathbf{R}_{\mathbf{j}} = -\mathbf{F}_{\mathbf{j}}$

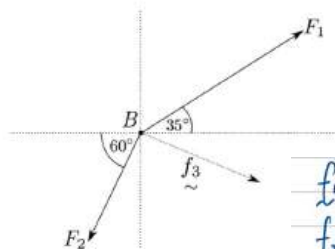
$$\Rightarrow \mathbf{M} = 20\sqrt{3} - 20$$

$$= 20(\sqrt{3} - 1)\text{ N}$$



A25	ASCHAM 2023	☒	☐
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Two forces are pulling on a boat at B and their sum (resultant) is shown as vector $\vec{f}_3$. The two forces have magnitudes F_1 and F_2 respectively, and their angles with the horizontal are shown below. Which is the correct expression for $\vec{f}_3$?



A) $\vec{f}_3 = \left(F_1 \cos 35^\circ - \frac{F_2}{2} \right) \hat{i} + \left(F_1 \sin 35^\circ - \frac{\sqrt{3}F_2}{2} \right) \hat{j}$

B) $\vec{f}_3 = \left(\frac{F_2}{2} - F_1 \cos 35^\circ \right) \hat{i} + \left(\frac{\sqrt{3}F_2}{2} - F_1 \sin 35^\circ \right) \hat{j}$

C) $\vec{f}_3 = \left(F_1 \sin 35^\circ - \frac{\sqrt{3}F_2}{2} \right) \hat{i} + \left(F_1 \cos 35^\circ - \frac{F_2}{2} \right) \hat{j}$

D) $\vec{f}_3 = \left(\frac{\sqrt{3}F_2}{2} - F_1 \sin 35^\circ \right) \hat{i} + \left(\frac{F_2}{2} - F_1 \cos 35^\circ \right) \hat{j}$

$$\vec{f}_1 = F_1 \cos 35^\circ \hat{i} + F_1 \sin 35^\circ \hat{j}$$

$$\vec{f}_2 = -F_2 \cos 60^\circ \hat{i} - F_2 \sin 60^\circ \hat{j}$$

$$= -\frac{F_2}{2} \hat{i} - \frac{\sqrt{3}F_2}{2} \hat{j}$$

$$\vec{f}_3 = \vec{f}_1 + \vec{f}_2 = \left(F_1 \cos 35^\circ - \frac{F_2}{2} \right) \hat{i} + \left(F_1 \sin 35^\circ - \frac{\sqrt{3}F_2}{2} \right) \hat{j}$$

A26	HOMEBUSH 2021	☒	☐
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Two forces are acting on an object: 200 N acting at $N65^\circ W$ and 250 N acting at $N30^\circ E$. 3

Calculate the resultant force $\vec{F}$ acting on the object, correct to one decimal place.

$$\vec{F}_1 = -200 \cos 25^\circ \hat{i} + 200 \sin 25^\circ \hat{j}$$

$$\vec{F}_2 = 250 \cos 60^\circ \hat{i} + 250 \sin 60^\circ \hat{j}$$

$$\vec{F} = \vec{F}_1 + \vec{F}_2$$

$$= -200 \cos 25^\circ \hat{i} + 200 \sin 25^\circ \hat{j} + 250 \cos 60^\circ \hat{i} + 250 \sin 60^\circ \hat{j}$$

$$= -56.26 \hat{i} + 801.03 \hat{j}$$

$$|\vec{F}| = \sqrt{56.26^2 + 801.03^2}$$

$$= 306.2 \text{ N (1 dp)}$$

$$\tan \theta = \frac{801.03}{56.26} = 5.3504$$

$$\theta = 79.4^\circ$$

$$N 10.6^\circ W$$

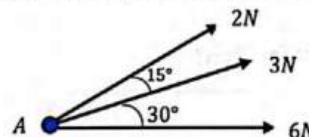
A27	KNOX 2022	☒	☐
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Forces of 6 N, 3 N and 2 N act on an object, considered point A, as shown in the diagram.

The vector sum of these forces acting on the object at point A is called the resultant force. The force 6 N is acting along a horizontal plane.

(i) Find the magnitude of the resultant force, correct to one decimal place.

(ii) Find the direction of this resultant force, correct to one decimal place.



Resolve forces into horizontal and vertical components.

For horizontal components, the sum gives:

$$6 + 3 \cos 30^\circ + 2 \cos 45^\circ = 10.0122... \text{ N}$$

For vertical components, perpendicular to force 6 N, the sum gives:

$$3 \sin 30^\circ + 2 \sin 45^\circ = 2.9142... \text{ N}$$

$$\therefore \text{magnitude of the resultant force} = \sqrt{(10.01)^2 + (2.91)^2} = 10.4 \text{ N}$$

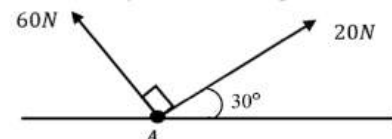
To find the direction of the resultant force which is a vector, we need the angle, θ , it makes with the horizontal.

$$\therefore \tan \theta = \frac{2.9142...}{10.0122...}$$

$$\theta = 16.2^\circ$$

A28	MANLY 2023	☒	☐
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Forces of 60 N and 20 N act on an object, considered point A, as shown in the diagram.



(i) Find the sum of the resolved parts of the forces in the horizontal direction.

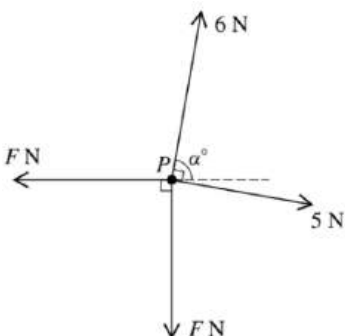
(ii) The vector sum of these forces acting on the object at point A is called the resultant force.

Find the resultant force vector in the form $x\hat{i} + y\hat{j}$.

i	$20 \cos 30^\circ + 60 \sin 120^\circ = 10\sqrt{3} - 30$ newtons
ii	vertical direction $20 \sin 30^\circ + 60 \sin 120^\circ$ $= 10 + 30\sqrt{3}$ $\therefore (10\sqrt{3} - 30)\hat{i} + (10 + 30\sqrt{3})\hat{j}$



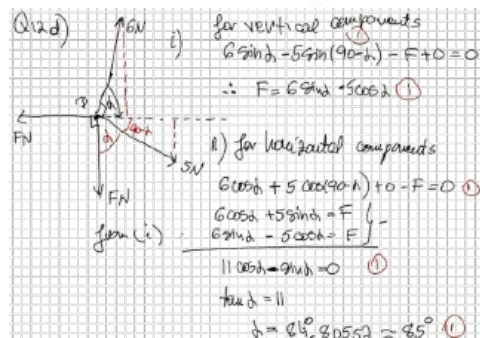
A29	NORTH SYDNEY BOYS 2023	☒	☐
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A particle P is in equilibrium under the action of four forces of magnitude 6 N, 5 N, F N and F N acting in the directions shown in the diagram on the right.

(i) By considering the vertical components of the forces, find an expression for F in terms of α .

(ii) Hence, determine the value of α , correct to the nearest degree.



A30	KILLARA 2022	☒	☐
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The diagram shows the force of gravity, W , that pulls straight down on an object. Two forces of 200 N and 300 N are exerted by cables as shown.

If the resultant force is zero, find θ and W .

Equating up and down and left and right

$$200 \sin 40^\circ + 300 \sin \theta - |W| = 0 \quad (1)$$

$$300 \cos \theta - 200 \cos 40^\circ = 0 \quad (2)$$

$$300 \cos \theta = 200 \cos 40^\circ$$

$$\theta = \cos^{-1} \left(\frac{2}{3} \cos 40^\circ \right)$$

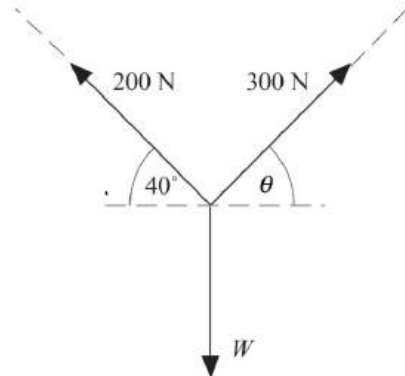
$$\theta = 59.28 \dots$$

$$\text{sub } \theta = 59.28 \dots \text{ into (1)}$$

$$200 \sin 40^\circ + 300 \sin 59.28 \dots = W$$

$$|W| = -386.49 \dots$$

$$\therefore \theta = 59.28 \dots \text{ and } W = -386.49 \dots$$



A31	RIVERVIEW 2022	☒	☐
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The diagram shows three forces acting on an object. Find the magnitude of the resultant force in Newton and its direction to the nearest degree.

b. Method: write each vector in the form of $a\mathbf{i} + b\mathbf{j}$ and then add for Resultant

$$40 \text{ N} : 40 \cos 150^\circ \mathbf{i} + 40 \sin 150^\circ \mathbf{j} = -20\sqrt{3} \mathbf{i} + 20 \mathbf{j}$$

$$30 \text{ N} : 30 \cos 210^\circ \mathbf{i} + 30 \sin 210^\circ \mathbf{j} = -15\sqrt{3} \mathbf{i} - 15 \mathbf{j}$$

$$20 \text{ N} : 20 \cos (-30^\circ) \mathbf{i} + 20 \sin (-30^\circ) \mathbf{j} = 10\sqrt{3} \mathbf{i} - 10 \mathbf{j}$$

$$\mathbf{R} = (-20\sqrt{3} - 15\sqrt{3} + 10\sqrt{3}) \mathbf{i} + (20 - 15 - 10) \mathbf{j}$$

$$= -25\sqrt{3} \mathbf{i} - 5 \mathbf{j}$$

Magnitude:

$$R : \sqrt{(-25\sqrt{3})^2 + (-5)^2} = \sqrt{1900}$$

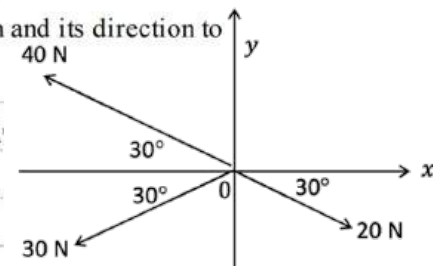
$$= 10\sqrt{19} \text{ N}$$

Direction: $\tan \theta = \frac{-5}{-25\sqrt{3}} = \frac{1}{5\sqrt{3}}$

$$\therefore \theta = \tan^{-1} \left(\frac{1}{5\sqrt{3}} \right) = 18.635^\circ$$

$$\approx 18.7^\circ$$

Hence, the resultant vector has magnitude $10\sqrt{19} \text{ N}$ and forms an angle of 18.7° with the positive direction of the x-axis.

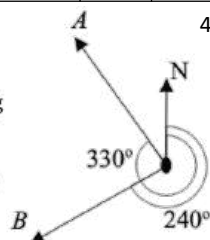




A32	FORT STREET 2021	☒	☐
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During a movie scene a camera is attached to three drones by inextensible chords, completely stretched horizontally, with no vertical movement. Drone A is applying a force of 30N on a bearing of 330° , while drone B is applying a force of 36N on a bearing of 240° , as shown in the diagram.

If the camera is to be kept stationary, calculate the bearing of drone C and the force it needs to apply to the camera, giving your answers to 2 decimal places.



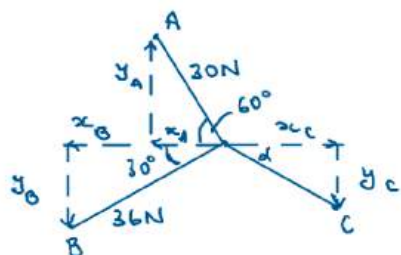
$$\begin{aligned}\underline{A} &= x_A \underline{i} + y_A \underline{j} \\ &= -30 \cos 60^\circ \underline{i} + 30 \sin 60^\circ \underline{j}\end{aligned}$$

$$= -15 \underline{i} + 15\sqrt{3} \underline{j}$$

$$\begin{aligned}\underline{B} &= x_B \underline{i} + y_B \underline{j} \\ &= -36 \cos 30^\circ \underline{i} + 36 \sin 30^\circ \underline{j}\end{aligned}$$

$$= -18\sqrt{3} \underline{i} + 18 \underline{j}$$

$$\underline{C} = x_C \underline{i} + y_C \underline{j}$$



Since the camera is not moving:

$$\underline{A} + \underline{B} + \underline{C} = 0$$

$$(-15 - 18\sqrt{3} + x_C) \underline{i} + (15\sqrt{3} - 18 + y_C) \underline{j} = 0 \underline{i} + 0 \underline{j}$$

$$\Rightarrow x_C = 15 + 18\sqrt{3} \quad ; \quad y_C = 18 - 15\sqrt{3}$$

$$\begin{aligned}\text{Therefore } |\underline{C}| &= \sqrt{x_C^2 + y_C^2} \\ &= \sqrt{2196}\end{aligned}$$

$$= 46.86149 \dots$$

$$\approx 46.86 \text{ N (2 d.p.)}$$

$$\alpha = \tan^{-1} \left(\frac{18 - 15\sqrt{3}}{15 + 18\sqrt{3}} \right)$$

$$= -9.80557 \dots$$

$$\approx -9.81^\circ \text{ (2 d.p.)}$$

Therefore drone C must apply a force of 46.86 N on a bearing of 99.81° .

A33	NORTH SYDNEY BOYS 2021	☒	☐
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The diagram below shows a mass being acted upon by a number of forces whose magnitudes are labelled. All forces are in newtons, and the system is in equilibrium.

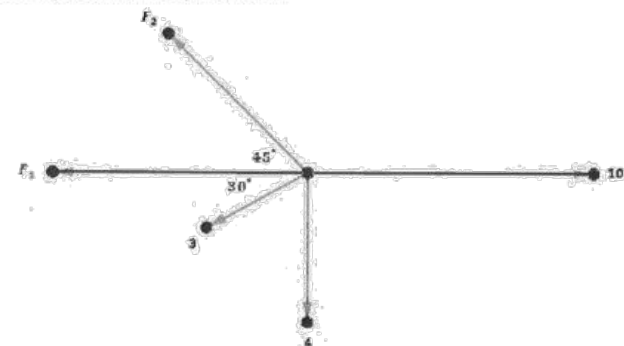
The value of F_2 is

(A) $\frac{\sqrt{2}}{2} (8 + 3\sqrt{3})$

(B) $\frac{11\sqrt{2}}{2}$

(C) $\frac{3\sqrt{2}}{2}$

(D) 7



A34	NORTH SYDNEY BOYS 2021	☒	☐
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Kylie drives her car from the origin all along the vector $\overrightarrow{OA} = \begin{pmatrix} 6 \\ 5 \end{pmatrix}$. Then from A she drives all along the vector $\overrightarrow{AB} = \begin{pmatrix} -3 \\ 3 \end{pmatrix}$ and after that, from B she drove to C along the vector $\overrightarrow{BC} = \begin{pmatrix} p \\ p+2 \end{pmatrix}$, where p is a positive constant.

Given that the vector position $\overrightarrow{OC}$ from her starting to her final position has a magnitude of 13, what is the value of p ?

A. 5

B. 2

C. 3

D. 1

7. The vector $\overrightarrow{OC} = \overrightarrow{OA} + \overrightarrow{AB} + \overrightarrow{BC}$
 $= 6\underline{i} + 5\underline{j} - 3\underline{i} + 3\underline{j} + p\underline{i} + (p+2)\underline{j}$

$$= (3+p)\underline{i} + (10+p)\underline{j}$$

Now, as the magnitude of $\overrightarrow{OC}$ is 13 then

$$\sqrt{(3+p)^2 + (10+p)^2} = 13$$

By squaring both sides, we get

$$(3+p)^2 + (10+p)^2 = 169$$

$$9 + 6p + p^2 + 100 + 20p + p^2 = 169$$

$$2p^2 + 26p - 60 = 0$$

Dividing by 2, we get

$$p^2 + 13p - 30 = 0$$

$$(p+15)(p-2) = 0$$

$$p = -15 \text{ or } p = 2$$

As p is a positive constant then $p = 2$.

Hence, the correct option is B.

B1	PENRITH 2022	☒	☐
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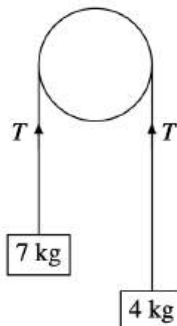
The diagram below shows a pulley with two objects of mass 7 kg and 4 kg attached to the ends of a light, inelastic string that passes through a smooth pulley. 1

The tension in the string is T newtons.

Take $g = 9.8 \text{ m}\cdot\text{s}^{-2}$

The downwards acceleration of the 7 kg, to 2 significant figures, is:

- A. $2.7 \text{ m}\cdot\text{s}^{-2}$
 B. $9.8 \text{ m}\cdot\text{s}^{-2}$
 C. $11 \text{ m}\cdot\text{s}^{-2}$
 D. $35 \text{ m}\cdot\text{s}^{-2}$

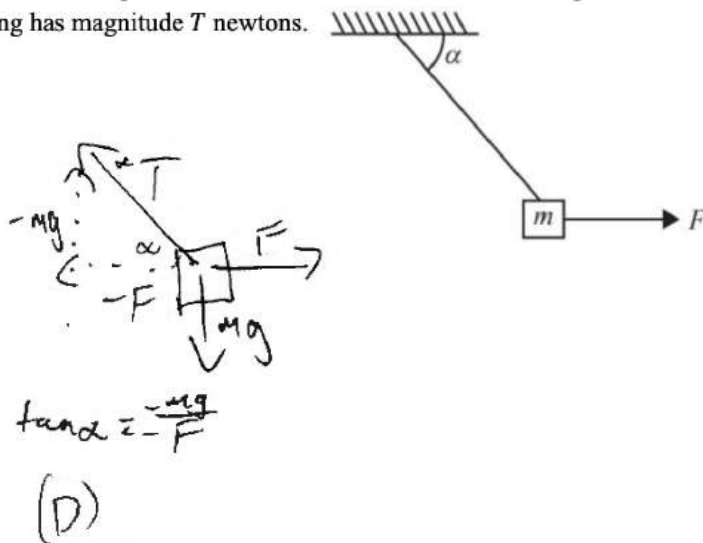


B2	KILLARA 2021	☒	☐
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A particle of mass m kilograms hangs from a string that is attached to a fixed point. The particle is acted on by a horizontal force of magnitude F newtons. The system is in equilibrium when the string makes an angle α to the horizontal, as shown in the diagram below. The tension in the string has magnitude T newtons. 1

The value of $\tan(\alpha)$ is

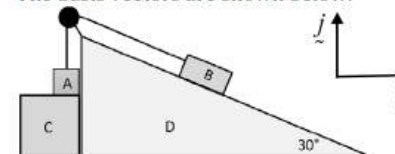
- (A) $\frac{mg}{T}$
 (B) $\frac{T}{mg}$
 (C) $\frac{T}{F}$
 (D) $\frac{mg}{F}$



B3	JAMES RUSE 2020	☒	☐
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The diagram below shows object A of mass 8 kg and object B of mass 5 kg. Both objects are attached to the ends of a light inextensible string that passes over a smooth pulley. Object C is fixed to the ground, and D is a fixed plane at an inclination of 30° degrees as shown. All objects are at rest, and all surfaces are smooth. g is $10 \text{ m}\cdot\text{s}^{-2}$.

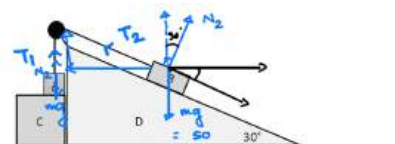
The basis vectors are shown below.



Let N_1 be the normal force of object C acting on object A and let N_2 be the normal force of the inclined plane acting on object B . Let T_1 and T_2 be the tension forces in the string applied to object A and object B , respectively.

Given $\begin{vmatrix} T_1 \\ T_2 \end{vmatrix} = t$ Newtons and $N_2 = \begin{bmatrix} \frac{n}{2} \\ \frac{\sqrt{3}}{2}n \end{bmatrix}$

- (i) Express T_2 in terms of t in column vector form.
 (ii) By setting up vector equations, find the values of t and n .
 (iii) Hence, find the components of N_1 .



$$\begin{aligned} |T_1| &= |T_2| = t \\ T_2 &= -|T_2| \cos 30^\circ \hat{i} + |T_2| \sin 30^\circ \hat{j} \\ &= -t \times \frac{\sqrt{3}}{2} \hat{i} + t \times \frac{1}{2} \hat{j} \\ &= -\frac{\sqrt{3}t}{2} \hat{i} + \frac{t}{2} \hat{j} = \begin{pmatrix} -\frac{\sqrt{3}t}{2} \\ \frac{t}{2} \end{pmatrix} \end{aligned}$$

Since B is at rest:

$$\begin{aligned} \text{Vert. comp. of } T_2 + \text{Vert. comp. of } N_2 &= mg \\ \frac{t}{2} + |N_2| \times \cos 30^\circ &= 5 \times 10 \\ \frac{t}{2} + n \times \frac{\sqrt{3}}{2} &= 50 \quad \text{--- (1)} \end{aligned}$$

Also, hor. comp. of $T_2 =$ hor. comp. of N_2 .

$$\begin{aligned} \frac{\sqrt{3}}{2}t &= |N_2| \cos 60^\circ \\ \frac{\sqrt{3}}{2}t &= \frac{n}{2} \\ t &= \frac{n}{\sqrt{3}} \quad \text{--- (2)} \\ \frac{n}{2\sqrt{3}} + \frac{\sqrt{3}n}{2} &= 50 \\ \frac{n + 3n}{2\sqrt{3}} &= 50 \\ 4n &= 50 \times 2\sqrt{3} \\ n &= 25\sqrt{3} \\ \therefore t &= 25 \end{aligned}$$

(iii).

$$\begin{aligned} T_1 + N_1 &= mg \\ t + N_1 &= 80 \\ N_1 &= 80 - 25 \\ N_1 &= 55 \\ \therefore N_1 &= \begin{pmatrix} 0 \\ 55 \end{pmatrix} \end{aligned}$$

B4	JAMES RUSE 2024	☒	☐
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Consider the two-body construction shown in the diagram below. A crate, having mass $m_1 = 2500$ kg, lies on a smooth, inclined plane. It is connected by a light, inextensible cable through a smooth pulley to a second crate having mass $m_2 = 4000$ kg. The plane has an angle of inclination of 30° .

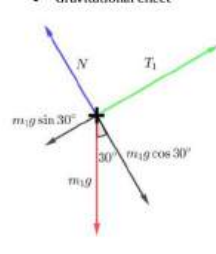
Taking the upward direction of the incline as positive, find the acceleration of m_1 in terms of g in its simplest form.

On m_1 , we have:

On m_2 , we have:

- Tension
- Normal reaction force
- Gravitational effect

- Tension
- Gravitational effect



For mass 2, basis vectors for forces will be as:

- Positive horizontal, left-to-right.
- Positive vertical, downward (since we are anticipating a certain direction given the setup, but it is not critical to make the downward direction positive).

Then, resolving forces on mass 2:

$$m_2g - T_2 = m_2a_{2\text{ve}} \quad \dots (2)$$

$$0 = m_2a_{2\text{horiz}} \rightarrow a_{2\text{horiz}} = 0$$

Now, since the cable is inextensible, the tension T_1 on mass 1 exerted by the cable is the same (in magnitude) as that exerted on mass 2, T_2 . Hence

$$T_1 = T_2$$

and the accelerations (their magnitudes) of the masses must be the same (if not, the cable would compress or break, depending on which mass has the greater acceleration). This implies $a_{1\text{horiz}} = a_{2\text{vert}} = a$.

Hence, we have from (1) and (2):

$$T - \frac{1}{2}m_1g = m_1a \quad \dots (3)$$

$$m_2g - T = m_2a \quad \dots (4)$$

Add (3) and (4):

$$(m_2 - \frac{1}{2}m_1)g = (m_1 + m_2)a$$

and so

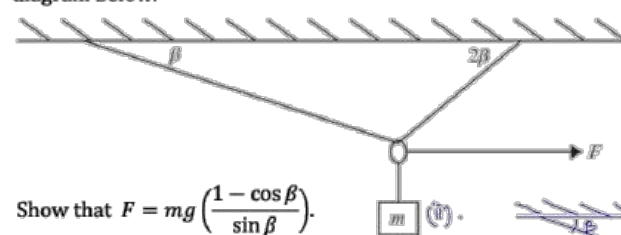
$$a = \frac{(2m_2 - m_1)}{2(m_1 + m_2)}g = \frac{11}{26}g$$

given $m_1 = 2500$ kg, $m_2 = 4000$ kg.

B5	CHERRYBROOK TECH 2022	☒	☐
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(i) Show that $\frac{\cos \beta - \cos 2\beta}{\sin \beta + \sin 2\beta} = \frac{1 - \cos \beta}{\sin \beta}$

(ii) A light inextensible string is connected at each end to a horizontal ceiling. A mass of m kilograms hangs from a smooth ring on the string. A horizontal force of F newtons is applied to the string. The tension in the string has a constant magnitude and the system is in equilibrium. At one end string makes an angle β with the ceiling and at the other end the string makes an angle 2β , as shown in the diagram below.



Show that $F = mg \left(\frac{1 - \cos \beta}{\sin \beta} \right)$.

(i) L.H.S

$$\frac{\cos \beta - \cos 2\beta}{\sin \beta + \sin 2\beta} = \frac{\cos \beta - (2\cos^2 \beta - 1)}{\sin \beta + 2\sin \beta \cos \beta}$$

$$= \frac{\cos \beta - 2\cos^2 \beta + 1}{\sin \beta (1 + 2\cos \beta)}$$

$$= \frac{-2\cos^2 \beta + \cos \beta + 1}{\sin \beta (1 + 2\cos \beta)}$$

$$= \frac{-2\cos^2 \beta + 2\cos \beta - \cos \beta + 1}{\sin \beta (1 + 2\cos \beta)}$$

$$= \frac{-2\cos \beta (\cos \beta - 1) - 1(\cos \beta - 1)}{\sin \beta (1 + 2\cos \beta)}$$

$$= \frac{-(2\cos \beta - 1)(\cos \beta - 1)}{\sin \beta (1 + 2\cos \beta)}$$

$$= -\frac{(2\cos \beta - 1)(\cos \beta - 1)}{\sin \beta (1 + 2\cos \beta)}$$

$$= -\frac{(\cos \beta - 1)}{\sin \beta} = \frac{1 - \cos \beta}{\sin \beta}$$

Equating the horizontal components

$$T \sin 2\beta + F = T \cos \beta$$

$$\therefore F = T \cos \beta - T \sin 2\beta$$

$$F = T(\cos \beta - \sin 2\beta) \rightarrow (1)$$

Equating the vertical components

$$T \sin \beta + T \sin 2\beta = mg$$

$$\text{or } T = \frac{mg}{\sin \beta + \sin 2\beta} \rightarrow (2)$$

Substituting the value of T from (2) into (1)

$$F = \frac{mg(\cos \beta - \sin 2\beta)}{(\sin \beta + \sin 2\beta)}$$

$$= mg \left(\frac{\cos \beta - \cos 2\beta}{\sin \beta + \sin 2\beta} \right)$$

Using the result from part (i)

$$F = mg \left(\frac{1 - \cos \beta}{\sin \beta} \right)$$



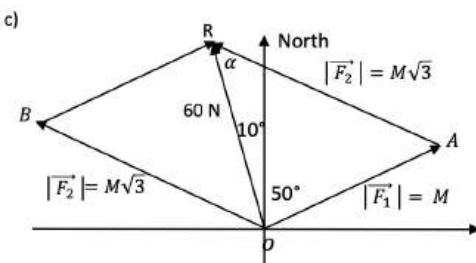
B6	S&G 2022	☒	☐
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The resultant force on a particle, which is moving subject to the forces $\vec{F}_1$ and $\vec{F}_2$, has magnitude 60N and its direction is on a bearing of 350° .

The force $\vec{F}_1$ has magnitude M and its direction is on a bearing of 50° .

The force $\vec{F}_2$ has magnitude $M\sqrt{3}$.

Calculate the magnitude and direction of the force $\vec{F}_2$.



The diagram shows the forces $\vec{F}_1$, $\vec{F}_2$ and their resultant vector of $\vec{OR}$ which is the diagonal of the parallelogram OARB formed by the vector sum of the two forces.

From the geometry of the figure, we have angle $\angle ROA$ equals $10^\circ + 50^\circ = 60^\circ$.

Using the cosine rule in triangle OAR, we get

$$\begin{aligned}\cos 60^\circ &= \frac{M^2 + 60^2 - (M\sqrt{3})^2}{2 \times M \times 60} \\ \frac{1}{2} &= \frac{M^2 + 3600 - 3M^2}{120M} \\ \frac{1}{2} &= \frac{3600 - 2M^2}{120M} \\ \frac{1}{2} &= \frac{1800 - M^2}{60M}\end{aligned}$$

$$60M = 3600 - 2M^2$$

$$2M^2 + 60M - 3600 = 0$$

$$M^2 + 30M - 1800 = 0$$

$$(M - 30)(M + 60) = 0$$

$$\text{So } M = 30 \text{ or } -60$$

As the magnitude of a force must positive then $M > 0$ then $M = 30$.

Hence, the magnitude of $|\vec{F}_2| = 30\sqrt{3} \text{ N}$

Now, let $\angle ORA = \alpha$.

Using the sine rule in triangle OAR, we get

$$\frac{\sin \alpha}{30} = \frac{\sin 60^\circ}{30\sqrt{3}}$$

$$30\sqrt{3} \sin \alpha = 30 \sin 60^\circ$$

$$\sqrt{3} \sin \alpha = \sin 60^\circ$$

$$\sin \alpha = \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{3}} = \frac{1}{2}$$

So, $\alpha = 30^\circ$ or $\alpha = 150^\circ$. But α cannot be 150° as it makes the angle sum of triangle OAR more than 180° . This means $\alpha = 30^\circ$.

$\angle ROB = 30^\circ$ (Alternate angles on parallel lines are equal)

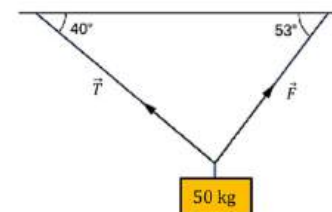
This indicates that $\angle NOB = 30^\circ + 10^\circ = 40^\circ$

Hence, the bearing of $\vec{F}_2$ is $360^\circ - 40^\circ = 320^\circ$.

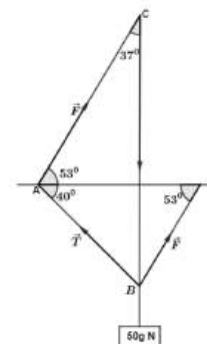
and from above the magnitude of $|\vec{F}_2|$ is $30\sqrt{3} \text{ N}$.

B7	HURLSTONE 2023	☒	☐
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A 50 kg weight is hung by a cable so that the two portions of the cable make angles of 40° and 53° , respectively, with the horizontal as shown below.



Find the magnitudes of the tension forces $\vec{T}$ and $\vec{F}$ in the cables if the resultant force acting on the object is zero. Give your answer in Newtons by rounding your answer to two decimal places. (use $g = 9.8$)



$$50 \text{ kg} = 50g \text{ N}$$

Representing the forces in a triangle as shown above and using Sine rule in triangle ABC

$$\frac{|\vec{T}|}{\sin 37^\circ} = \frac{|\vec{F}|}{\sin 50^\circ} = \frac{50g}{\sin (53^\circ + 40^\circ)}$$

$$\frac{|\vec{T}|}{\sin 37^\circ} = \frac{|\vec{F}|}{\sin 50^\circ} = \frac{50g}{\sin 93^\circ}$$

$$\frac{|\vec{T}|}{\sin 37^\circ} = \frac{50g}{\sin 93^\circ} \quad \frac{|\vec{F}|}{\sin 50^\circ} = \frac{50g}{\sin 93^\circ}$$

$$|\vec{T}| = \frac{50g}{\sin 93^\circ} \times \sin 37^\circ = 295.29 \text{ N}$$

$$|\vec{F}| = \frac{50g}{\sin 93^\circ} \times \sin 50^\circ = 375.89 \text{ N}$$

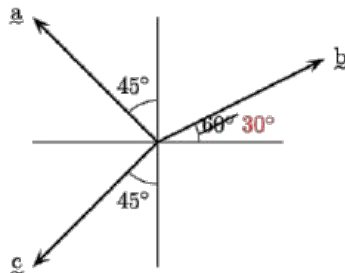


B8

NORMANHURST BOYS 2021



A game of tug-of-war involves three teams who each attach their rope to a weight to pull in their direction as shown. Let $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ be the pulling force of Team A, Team B and Team C respectively.



Correction: the angle 60° was corrected to 30° .

It is given that $|\mathbf{c}| = 200$ N. All teams are unable to move the weight from its starting position in any direction.

i. Show that

$$\mathbf{a} = \left(-\frac{1}{\sqrt{2}}|\mathbf{a}|\right)\mathbf{i} + \left(\frac{1}{\sqrt{2}}|\mathbf{a}|\right)\mathbf{j} \quad \text{and} \quad \mathbf{b} = \left(\frac{\sqrt{3}}{2}|\mathbf{b}|\right)\mathbf{i} + \left(\frac{1}{2}|\mathbf{b}|\right)\mathbf{j}$$

ii. Briefly explain why $\mathbf{a} + \mathbf{b} + \mathbf{c} = \mathbf{0}$.

iii. Hence find the exact value of $|\mathbf{a}|$ and $|\mathbf{b}|$.

Equation (14.4) + (14.5):

$$\frac{1}{2}|\mathbf{b}|(\sqrt{3} + 1) - 200\sqrt{2} = 0$$

$$|\mathbf{b}| = \frac{400\sqrt{2}}{\sqrt{3} + 1}$$

$$= \frac{400\sqrt{2}(\sqrt{3} - 1)}{(\sqrt{3})^2 - 1^2}$$

$$= \frac{400(\sqrt{6} - \sqrt{2})}{2}$$

$$\therefore |\mathbf{b}| = 200(\sqrt{6} - \sqrt{2}) \text{ N} \quad (14.6)$$

Substitute equation (14.6) into (14.4):

$$-\frac{1}{\sqrt{2}}|\mathbf{a}| + \frac{\sqrt{3}}{2} \times 200(\sqrt{6} - \sqrt{2}) - 100\sqrt{2} = 0$$

$$\frac{1}{\sqrt{2}}|\mathbf{a}| = 100(\sqrt{18} - \sqrt{6}) - 100\sqrt{2}$$

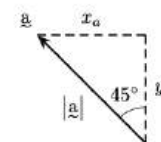
$$|\mathbf{a}| = 100\sqrt{2}(2\sqrt{2} - \sqrt{6})$$

$$\therefore |\mathbf{a}| = 100(4 - 2\sqrt{3}) \text{ N}$$

Corrected solution for Question 14 (b)
(Bhamra)

b. i. (2 marks)

- ✓ [1] for component form of $\mathbf{a}$
- ✓ [1] for component form of $\mathbf{b}$



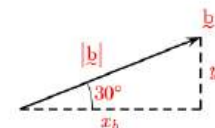
$$x_a = -|\mathbf{a}| \cos 45^\circ$$

$$= -\frac{1}{\sqrt{2}}|\mathbf{a}|$$

$$y_a = \frac{1}{\sqrt{2}}|\mathbf{a}|$$

$$= \frac{1}{\sqrt{2}}|\mathbf{a}|$$

$$\therefore \mathbf{a} = \left(-\frac{1}{\sqrt{2}}|\mathbf{a}|\right)\mathbf{i} + \left(\frac{1}{\sqrt{2}}|\mathbf{a}|\right)\mathbf{j}$$



$$x_b = |\mathbf{b}| \cos 30^\circ$$

$$= \frac{\sqrt{3}}{2}|\mathbf{b}|$$

$$y_b = |\mathbf{b}| \sin 30^\circ$$

$$= \frac{1}{2}|\mathbf{b}|$$

$$\therefore \mathbf{b} = \left(\frac{\sqrt{3}}{2}|\mathbf{b}|\right)\mathbf{i} + \left(\frac{1}{2}|\mathbf{b}|\right)\mathbf{j}$$

ii. (1 mark)

- ✓ [1] for correct explanation

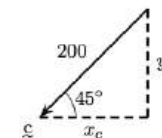
The forces of vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are in equilibrium as the weight's motion remains constant at rest.

Hence the resultant force is zero:

$$\mathbf{a} + \mathbf{b} + \mathbf{c} = \mathbf{0}$$

iii. (4 marks)

- ✓ [1] for component form of $\mathbf{c}$
- ✓ [1] for simultaneous equations
- ✓ [1] for correct values of $|\mathbf{a}|$
- ✓ [1] for correct values of $|\mathbf{b}|$



$$x_c = -200 \cos 45^\circ$$

$$= -100\sqrt{2}$$

$$y_c = -200 \sin 45^\circ$$

$$= -100\sqrt{2}$$

$$\mathbf{c} = (-100\sqrt{2})\mathbf{i} + (-100\sqrt{2})\mathbf{j}$$

From part (ii):

$$\left(-\frac{1}{\sqrt{2}}|\mathbf{a}|\right) + \left(\frac{\sqrt{3}}{2}|\mathbf{b}|\right) + (-100\sqrt{2}) = 0$$

$$\therefore -\frac{1}{\sqrt{2}}|\mathbf{a}| + \frac{\sqrt{3}}{2}|\mathbf{b}| - 100\sqrt{2} = 0 \quad (14.4)$$

$$\frac{1}{\sqrt{2}}|\mathbf{a}| + \frac{1}{2}|\mathbf{b}| - 100\sqrt{2} = 0 \quad (14.5)$$



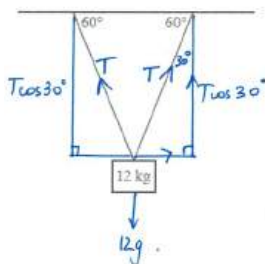
B9 SYDNEY GIRLS 2021



A 12 kg mass is suspended in equilibrium from a horizontal ceiling by two identical light strings. Each string makes an angle of 60° with the ceiling, as shown. If the force of gravity is $g \text{ m/s}^2$ then the magnitude, in newtons, of the tension in each string is equal to:

1

- (A) $12g$
(B) $24g$
(C) $4\sqrt{3}g$
(D) $8\sqrt{3}g$



$$\begin{aligned} 2T \cos 30^\circ &= 12g \\ T &= \frac{12g}{2 \cos 30^\circ} \\ T &= \frac{12g}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = 4\sqrt{3}g \end{aligned}$$